WEEK 9

Aim:

To develop a **College Event Registration UI** using **React (JSX – JavaScript Extension)** with a registration form and to manage and display user-entered information using React state and event handling.

Description:

This experiment demonstrates the development of a real-world user interface using **React and JSX**. The application provides a College Event Registration form where users can enter their name, email, phone number, and select their department. React's **useState** hook is used to store and manage the form data. A common event handler is used to update the form fields dynamically. When the user submits the form, **preventDefault()** is used to prevent the page from refreshing, and the submitted details are displayed below the form using conditional rendering. The application demonstrates JSX, components, state management, controlled form inputs, event handling, and dynamic rendering in React.

Program:

College Event Registration UI Using React and JSX

Program Code:

```
import { useState } from "react";

function App() {

  const [formData, setFormData] = useState({
    name: "",
    email: "",
    phone: "",
    department: ""
  });

  const handleChange = (e) => {
    setFormData({
      ...formData,
      [e.target.name]: e.target.value
    });
  };
```

```
const [submittedData, setSubmittedData] = useState(null);

const handleSubmit = (e) => {
  e.preventDefault();
  setSubmittedData(formData);
};

return (
  <div>
    <h1>College Event Registration</h1>

    <p>Register for upcoming college events</p> <br /><br />

    <form onSubmit={handleSubmit}>

      <label>Full Name:</label>
      <input
        type="text"
        name="name"
        placeholder="Enter your name"
        value={formData.name}
        onChange={handleChange}
      />
      <br /><br />

      <label>Email:</label>
      <input
        type="email"
        name="email"
        placeholder="Enter your email"
        value={formData.email}
        onChange={handleChange}
      />

      <br /><br />

      <label>Phone Number:</label>
      <input
```

```
  type="tel"
  name="phone"
  placeholder="Enter your phone number"
  value={formData.phone}
  onChange={handleChange}
/>

<br /><br />

<label>Department:</label>
<select
  name="department"
  value={formData.department}
  onChange={handleChange}
>
  <option value="">Select Department</option> <option
  value="CSE">CSE</option> <option
  value="CSM">CSM</option> <option
  value="ECE">ECE</option> <option
  value="EEE">EEE</option>
  <option value="IT">IT</option>
</select>

<br /><br />

<button type="submit">Register</button>

 {submittedData && (
  <div>
    <br />

    <p>Registered Successfully</p>

    <br />

    <p>Name : {submittedData.name}</p>

    <br />

    <p>Email : {submittedData.email}</p>
```

```
            <br />

            <p>Phone : {submittedData.phone}</p>

            <br />

            <p>Department : {submittedData.department}</p> </div>
        )}

      </form>
    </div>
  );
}
export default App;
```

Execution of the Program:

Method	Purpose
React Component	Creates the College Event Registration UI
useState()	Stores and manages form data
handleChange()	Updates the form data when the user enters values
handleSubmit()	Handles form submission preventDefault() Prevents the page from refreshing
Conditional Rendering	Displays submitted details after registration

Working:

1. The React application displays the **College Event Registration** form. 2. The user enters their **Full Name, Email, Phone Number**, and selects their **Department**.
3. The entered values are stored in the formData state object. 4. When the **Register** button

is clicked, the handleSubmit() function is executed.
5. preventDefault() prevents the browser from refreshing the page. 6. The form data is stored in submittedData.
7. The submitted details are displayed below the registration form.

Output:

College Event Registration Page:

The application initially displays the registration form with fields for Full Name, Email, Phone Number, Department, and a Register button.

Registration Form:

The user enters the required details and selects a department from the dropdown. **Successful**

Registration:

After clicking the **Register** button, the message **"Registered Successfully"** is displayed along with the entered user details.

Output Screenshots:

Output 1: College Event Registration form before entering details.



Output 2: College Event Registration form after entering details with submitted details



Result:

The **College Event Registration UI** was successfully developed using **React and JSX**. The application demonstrates the use of JSX, **useState**, controlled form inputs, event handling, form submission using **preventDefault()**, and conditional rendering to dynamically display the submitted registration details.